Equilibrium

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Contents

1 Reaction Rate & Equilibrium

Core Concept 1.1: Reaction Rates

Atoms, ions, and molecules react with one another via collisions. The more frequent and energetic the collisions, the faster the reaction rate.

Take the formation of compound C as an example: $A + B \longrightarrow C$. The rate at which C forms depends on factors such as [A], [B], temperature, the presence of a catalyst, etc. and can be expressed mathematically as:

$$\text{Rate} = k[\text{A}]^m[\text{B}]^n$$

Here, k is the **rate constant**, which is specific to the given reaction and its conditions. m and n are the **partial reaction orders**, which represents a reactant's influence over the reaction rate. Here is a hypothetical plot of [A], [B], and [C] over time:

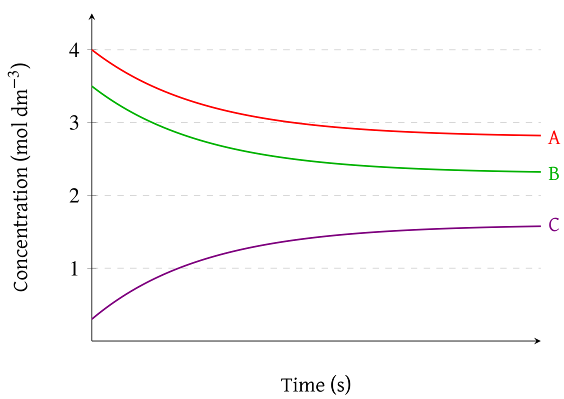

Figure 1: Equilibrium of $A + B \longrightarrow c$

In single-step reactions, partial reaction orders are equal to the stoichiometric coefficients of in the balanced chemical equation. However, in multi-step reactions, partial reaction orders may not directly correspond to the stoichiometric coefficients and must be determined experimentally.

Consider the decomposition of H_2O_2 ($H_2O_2 \longrightarrow H_2O + O_2$) as an example. When you graph the concentration of H_2O_2 over time, you get an exponential decay curve, where the demposition rate of H_2O_2 decreases as $[H_2O_2]$ decreases.

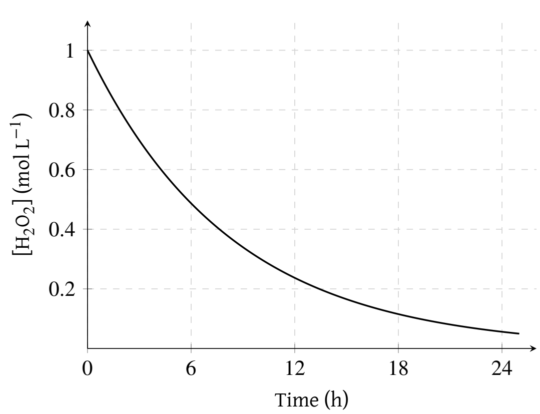

Figure 2: Decomposition of H_2O_2 over time

Core Concept 1.2: Equilibrium

Reactions do not always proceed in only one direction. Many reactions are **reversible**, meaning that the products can react to reform the reactants. When the rates of the forward and reverse reactions are equal, the system is said to be at **equilibrium**. Thus, for the general reversible shell reaction $AB \rightleftharpoons A + B$, the Eq reaction state can be expressed as:

$$k_{forward}[AB] = k_{backward}[A][B]$$

And the equilibrium constant K_c is defined as:

$$K_c = \frac{k_f}{k_b} = \frac{[A][B]}{[AB]}$$

Core Concept 1.3: Analyzing K_c

K_c provides insight into the position of equilibrium. A large K_c (>> 1) indicates product favorability, while a small K_c (<< 1) indicates reactant favorability.

Some examples of reactions that possess a **large** K_c (reaction to completion) include:

$Si + O_2 \longrightarrow SiO_2 \quad K_c \approx 10^{150}$

$2\,H_2 + O_2 \longrightarrow H_2O \quad K_c \approx 10^{81}$

Some examples of reactions that possess a **moderate** K_c (half-reaction) include:

$CO + \text{Hemoglobin} \longrightarrow CO \cdot \text{Hemoglobin} \quad K_c \approx 300$

$O_2 + \text{Hemoglobin} \longrightarrow O_2 \cdot \text{Hemoglobin} \quad K_c \approx 4$

$N_2O_4 \longrightarrow 2\,NO_2 \quad K_c \approx 0.2$

An example of a reaction that possess a **neglible** K_c (no reaction) include:

$N_2 + O_2 \longrightarrow 2\,NO \quad K_c \approx 10^{-31}$

The decomposition of Bromothymol Blue (*denoted as A*) exhibits a nice visual representation of reaction equilibria.

$HA \rightleftharpoons H^+ + A-$

If we start with a pure HA solution, the solution will gradually change color from yellow to green to blue as the forward reaction proceeds and $[A^-]$ rises. This particular compound is also known as an indicator, a concept covered in Core Concept 1.5.

Core Concept 1.4: Le Chatelier's Principle

Le Chatelier's Principle states that a system in equilibrium subjected to a stress will react in a direction that tends to reduce that stress and return back to Eq.

In more exact terms, stress will shift the **reaction quotient (Q)** — which represents the relative amounts of reactant and products at a given point in time before it has nessecarily reached equilibrium ($Q \neq K$) — and the system will react in a manner that shifts Q back to K.

$$Q = \frac{[A]_{current}[B]_{current}}{[AB]_{current}}$$

There are three major categories of stress in the context of Le Chatelier's Principle.

Core Concept 1.5: Le Chatelier's Principle (*Concentration Change*)

Shifting the point concentrations of any chemical participant will shift Q.

Let's reexamine the Bromothymol Blue reaction (HA $\rightleftharpoons$ H^+ + A-). If we add more H^+ to an initially green solution ([HA] = [A^-]), the system will increase the backwards reaction rate to consume excess H^+, resulting in a more yellow solution. Conversely, removing H^+ (*by adding* OH^-) will increase the forward reaction rate to produce more H^+, resulting in a more blue solution.

Consider another example ($PCl_5 \longrightarrow PCl_3 + Cl_2$) based on the visual below. The plot shows the abrupt addition of PCl_5 to a system originally at equilibrium. Consequently, $Q < K$ and the system reacts by increasing the forwards reaction rate to re-establish equilibrium, resulting in a subsequent increase in [PCl_3] and [Cl_2].

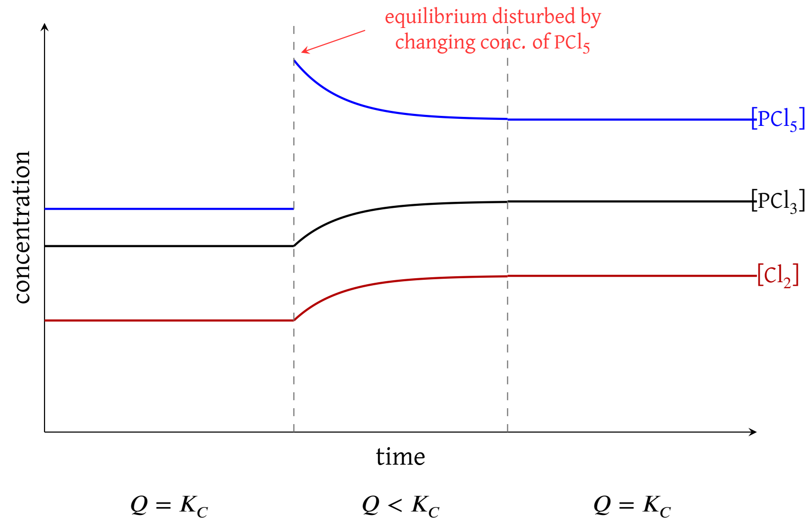

Figure 3: Disturbance of equilibrium by changing concentration of PCl_5

Core Concept 1.6: Le Chatelier's Principle (*Pressure & Volume Change*)

While concentration is the primary focus for solutions, pressure and volume are the key variables for gaseous systems. Increasing the pressure forces the equilibrium toward the side with fewer gas molecules to relieve the stress. Conversely, decreasing pressure allows the reaction to shift toward the side with more gas molecules.

The equilibrium constant in terms of partial pressures for gaseous reactions is written as K_p.

Consider the following reaction:

$$2\,NO_2(g) \rightleftharpoons N_2O_4(g) \qquad K_p = \frac{P_{N2O2}}{(P_{NO2})^2} \approx 7 \text{ at } 25°C$$

Increasing the volume (V) causes a drop in total pressure (P). To counteract this, the equilibrium shifts to the left, favoring the side with more moles of gas. Mathematically, this occurs because the partial pressure of NO_2 is squared in the denominator of the Q_p expression. A decrease in pressure impacts the squared term more significantly than the linear numerator, making $Q < K$ and forcing a shift toward the reactants.

Core Concept 1.7: Le Chatelier's Principle (*Temperature Change*)

A temperature shift is analogous to an increase or decrease of heat within a system, and the Q shift direction depends on whether heat is a net product (**exothermic**) or a reactant (**endothermic**) of the chemical reaction at hand. Reconsider the following gaseous reaction in the context of heat.

$2\,NO_2(g) \rightleftharpoons N_2O_4(g) + \text{heat}$

Much like the addition of N_2O_4, increasing the temperature drives the equilibrium to the left to consume the excess energy. Conversely, cooling the system shifts the reaction to the right as it attempts to regenerate the lost heat.

Chemical equilibrium in the context of temperature is eleaborated more upon in section 3.

2 Types of Equilibria

Equilibria can also be categorized into types.

Core Concept 2.1: Homogeneous Equilibria

In **homogeneous equilibria**, all reactants and products are in the same phase (solid, liquid, or gas), such as the following examples.

$Fe^{3+}(aq) + SCN^-(aq) \rightleftharpoons FeSCN^{2+}(aq)$ Solution

$H_2(g) + I_2(g) \rightleftharpoons 2\,HI(g)$ Gas

$HA(aq) \rightleftharpoons H^+(aq) + A^-(aq)$ Acid-Base

In **heterogeneous equilibria**, the reactants and products are in different phases and can be further subcategorized into three domains.

Core Concept 2.2: Heterogeneous Equilibria (*Solubility of Solids*)

The solubility of solids equilibrium refers to the dissociation and crystallization rate of solutes. Consider the dissolution of glucose. Solutes that do not dissociate are called **molecular or nonelectrolyte solutes**.

$C_6H_{12}O_6(s) \rightleftharpoons C_6H_{12}O_6(aq)$

Dissociation Rate = $k_f(SA)$

Crystallization Rate = $k_c(SA)[C_6H_{12}O_6(aq)]$

The dissociation rate only depends on the surface area (SA) of the solid glucose, while the crystallization rate depends on both the SA and $[C_6H_{12}O_6]$. At equilibrium:

$$K_{sp} = \frac{k_c(SA)}{k_f(SA)} = [C_6H_{12}O_6(aq)]$$

For glucose, the K_{sp} value is 5.1M at 25°C, meaning that the maximum $[C_6H_{12}O_6]$ in water is 5.1M. Beyond this concentration, any excess glucose will remain undissolved.

Now consider a different compound like NaCl. Solutes that dissociate are called **ionic or eletrolyte solutes**.

$NaCl(s) \rightleftharpoons Na^+(aq) + Cl^-(aq)$

Dissociation Rate = $k_f(SA)$

Crystallization Rate = $k_c(SA)[\text{Na}^+(\text{aq})][\text{Cl}^-(\text{aq})]$

At equilibrium:

$$K_{sp} = \frac{k_c(SA)}{k_f(SA)} = [\text{Na}^+(\text{aq})][\text{Cl}^-(\text{aq})]$$

At 25°C, the $K_{sp} \approx 37.3$, making the maximum [NaCl] in water approximately $6.11M$.

Core Concept 2.3: Heterogeneous Equilibria (*Solubility of Gases*) & Henry's Law

Gases can disolve into liquids just as how solids can. Consider the K_{sp} of CO_2 in water.

$CO_2(g) \rightleftharpoons CO_2(aq)$

When we reach equilibrium, we have the following:

$$K_H = \frac{[\text{CO}_2(\text{aq})]}{\text{P}_{\text{CO2}}} \rightarrow [\text{CO}_2(\text{g})] = K_H \text{P}_{\text{CO2}}$$

The equilibrium formula above is known as **Henry's Law**, and the constant K_H is called the **Henry's Constant** (M/atm), a quantitative measure of the proportion between a gas's solubility in a liquid and its partial pressure. The higher the K_H value, the easier it is for a gas to dissolve in a liquid.

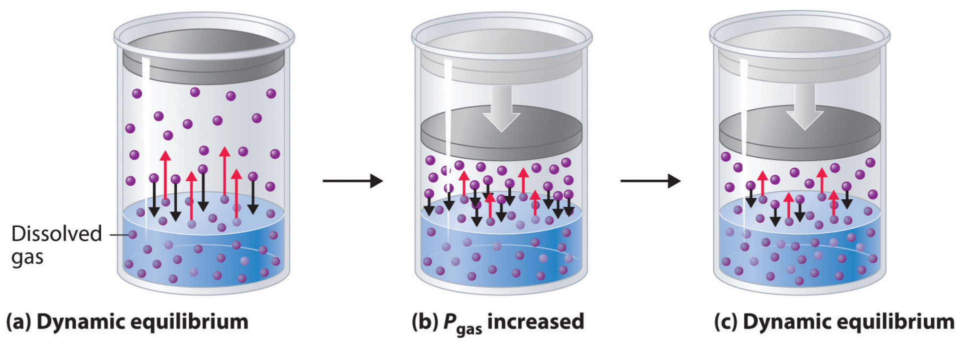

Figure 4: Gas solubility dynamic equilibrium

Core Concept 2.4: Heterogeneous Equilibria (*Phase Change*)

Phase changes also experience equilibriums. Consider the vaporization of water in a sealed flask.

$H_2O(l) \rightleftharpoons H_2O(g)$

Evaporation Rate = $k_f(SA)$

Condensation Rate = $k_c(SA)\ \text{P}_{\text{H}_2\text{O}}$

Just as the solubility equilibria for nonelectrolyte solutes is only dependent on the solvent concentration, the phase change equilibria for a gas to liquid transition is only dependent on $\text{P}_{\text{H}_2\text{O}}$.

$$K_{sp} = \frac{k_c(SA)}{k_f(SA)} = \text{P}_{\text{H}_2\text{O}}$$

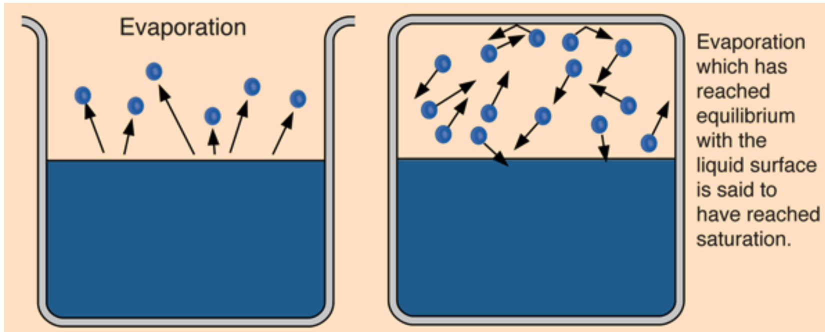

Figure 5: Phase change equilibrium of liquid-gas systems

Phase change equilibria become irrelevant for solid-liquid transitions because both pure solids and pure liquids have fixed, constant concentrations that are excluded from the equilibrium expression, leaving no variable quantities and rendering the equilibrium constant meaningless.

3 Shifting Equilibria

In **phase change equilibria**, the melting and boiling points of H_2O at 0°C and 100°C respectively are defined at a standard pressure of 1 atm; under varying pressure conditions, these transition temperatures become dynamic.

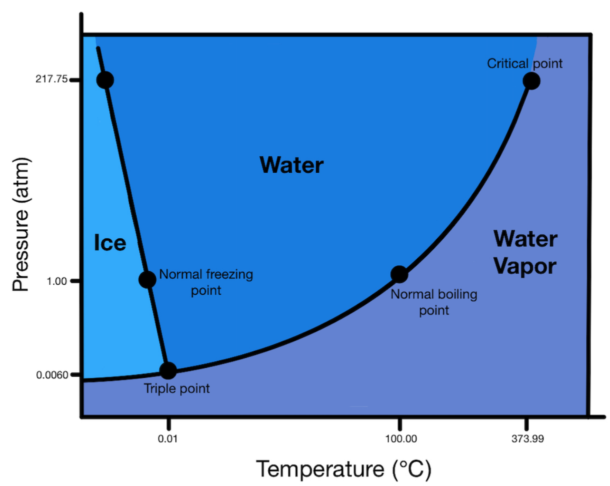

Figure 6: Phase diagram of water

Temperature shifts have a more nuanced effect in the context of equilibrium compared to the simple Le Chatelier's principle we have seen in Core Concept 1.7. Namely, the K_{sp} values of various solutes change across temperature ranges.

On the molecular scale, K_{sp} values of solid solutes increase with higher temperatures because the increased molecular motion of the solute and solvent molecules raises the probability of the solute entering the aqueous state.

In the context of gas solubility within a closed system, the Henry's Law constant K_H decreases with increasing temperature. Since gaseous molecules possess greater kinetic energy than their aqueous counterparts, the dissolution of a gas is an exothermic process; as temperature rises, the equilibrium shifts to favour the gaseous state, increasing partial pressure P at the expense of aqueous concentration C, thereby reducing $K_H = \frac{C}{P}$.

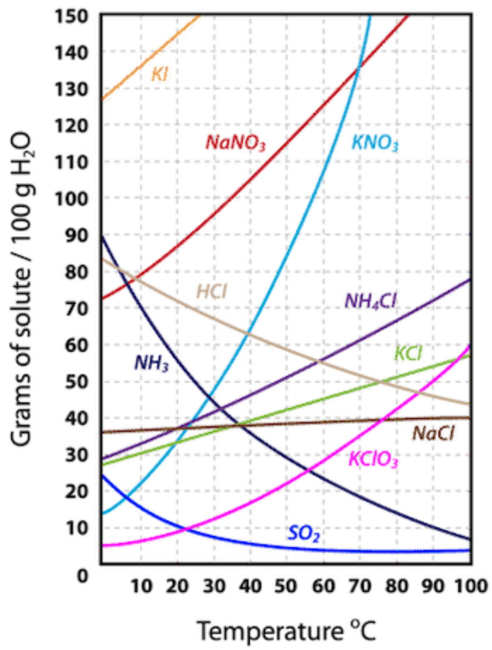

Figure 7: Solubility curve of various compounds